CS 2601 Linear and Convex Optimization

14. Dual LP

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Outline

- Dual LP
- Interpretation of dual problems
- Weak and strong duality
- Dual LP via Lagragian

Lower bounds in LP

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f(\boldsymbol{x}) = x_1 + 2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Any feasible solution $\boldsymbol{x}_0$ gives an upper bound on the optimal value f^*,

$$f^* \leq f(\boldsymbol{x}_0)$$

Can we also get a lower bound f_{LB} on f^*?

$$f^* \geq f_{\text{LB}}$$

Note. A lower bound on f^* is the same as a lower bound on $f(\boldsymbol{x})$ for all feasible $\boldsymbol{x} \in X$, i.e.

$$f^* \geq f_{\text{LB}} \iff f(\boldsymbol{x}) \geq f_{\text{LB}}, \quad \forall \boldsymbol{x} \in X$$

Lower bounds in LP (cont'd)

For any $\mu_1, \mu_2, \mu_3 \geq 0$,

$$
\begin{array}{lcrcrcrl}
\mu_1 \times & [& 2x_1 & + & x_2 & \geq & 2 &] \\
\mu_2 \times & [& x_1 & & & \geq & 0 &] \\
\mu_3 \times & [& & & x_2 & \geq & 0 &] \\
\hline
& & (2\mu_1 + \mu_2)x_1 & + & (\mu_1 + \mu_3)x_2 & \geq & 2\mu_1 =: \psi(\boldsymbol{\mu}) &
\end{array}
$$

We can set $2\mu_1 + \mu_2 = 1$ and $\mu_1 + \mu_3 = 2$ so the LHS becomes f.

Thus

$$f(\boldsymbol{x}) \geq \psi(\boldsymbol{\mu}) = 2\mu_1$$

for any $\boldsymbol{x} \in X$ and any μ_1, μ_2, μ_3 s.t.

$$2\mu_1 + \mu_2 = 1, \quad \mu_1 + \mu_3 = 2, \quad \mu_1, \mu_2, \mu_3 \geq 0$$

In particular, $f^* = \inf_{\boldsymbol{x} \in X} f(\boldsymbol{x}) \geq \psi(\boldsymbol{\mu})$ for such $\boldsymbol{\mu}$.

Lower bounds in LP (cont'd)

The quality of the lower bound $\psi(\boldsymbol{\mu}) = 2\mu_1$ varies for different $\boldsymbol{\mu}$.

- $\psi(0,1,2) = 0$, so

$$0 = \psi(0,1,2) \le f^* \le f(1,0) = 1$$

which also tells us $\boldsymbol{x}_0 = (1,0)$ is 1-suboptimal, i.e.

$$f(1,0) - f^* \le 1$$

- $\psi(\frac{1}{2},0,\frac{3}{2}) = 1$, so

$$1 = \psi(\frac{1}{2},0,\frac{3}{2}) \le f^* \le f(1,0) = 1$$

which tells us $f^* = 1$ and $\boldsymbol{x}_0 = (1,0)$ is the optimal solution!

Dual LP

To get the best lower bound, we maximize over μ_1, μ_2, μ_3,

primal LP	dual LP
$\min_{\boldsymbol{x} \in \mathbb{R}^2} \; f(\boldsymbol{x}) = x_1 + 2x_2$	$\max_{\boldsymbol{\mu} \in \mathbb{R}^3} \; \psi(\boldsymbol{\mu}) = 2\mu_1$
s.t. $2x_1 + x_2 \geq 2$	s.t. $2\mu_1 + \mu_2 = 1$
$x_1 \geq 0$	$\mu_1 + \mu_3 = 2$
$x_2 \geq 0$	$\mu_1 \geq 0$
	$\mu_2 \geq 0$
	$\mu_3 \geq 0$

The variables μ_1, μ_2, μ_3 are called dual variables.

We have one dual variable for each constraint in the primal problem.

Dual LP (cont'd)

Given $\boldsymbol{c} \in \mathbb{R}^n, \boldsymbol{A} \in \mathbb{R}^{k \times n}, \boldsymbol{b} \in \mathbb{R}^k, \boldsymbol{G} \in \mathbb{R}^{m \times n}, \boldsymbol{h} \in \mathbb{R}^m$, consider

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) = \boldsymbol{c}^T \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{G}\boldsymbol{x} \geq \boldsymbol{h} \end{aligned}$$

For $\boldsymbol{\lambda} \in \mathbb{R}^k$, $\boldsymbol{\mu} \in \mathbb{R}^m$ and $\boldsymbol{\mu} \geq \boldsymbol{0}$,

$$\boldsymbol{\lambda}^T \boldsymbol{A}\boldsymbol{x} + \boldsymbol{\mu}^T \boldsymbol{G}\boldsymbol{x} \geq \boldsymbol{\lambda}^T \boldsymbol{b} + \boldsymbol{\mu}^T \boldsymbol{h} =: \psi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

If $\boldsymbol{A}^T \boldsymbol{\lambda} + \boldsymbol{G}^T \boldsymbol{\mu} = \boldsymbol{c}$, then we can lower bound f^* by $f^* \geq \psi(\boldsymbol{\lambda}, \boldsymbol{\mu})$.

To maximize the lower bound, solve the following dual LP

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \psi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{b}^T \boldsymbol{\lambda} + \boldsymbol{h}^T \boldsymbol{\mu} \\ \text{s.t.} \quad & \boldsymbol{A}^T \boldsymbol{\lambda} + \boldsymbol{G}^T \boldsymbol{\mu} = \boldsymbol{c} \\ & \boldsymbol{\mu} \geq \boldsymbol{0} \end{aligned}$$

Dual LP (cont'd)

It is common to eliminate dual variables corresponding to $\boldsymbol{x} \geq \mathbf{0}$, and call the result the dual LP. Here are some common forms of dual LP,

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \boldsymbol{c}^T\boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \mathbf{0} \end{aligned}$$	$$\begin{aligned} \max_{\boldsymbol{y}} \quad & \boldsymbol{b}^T\boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^T\boldsymbol{y} \leq \boldsymbol{c} \end{aligned}$$
$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \boldsymbol{c}^T\boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b} \end{aligned}$$	$$\begin{aligned} \max_{\boldsymbol{y}} \quad & \boldsymbol{b}^T\boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^T\boldsymbol{y} = \boldsymbol{c} \\ & \boldsymbol{y} \geq \mathbf{0} \end{aligned}$$
$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \boldsymbol{c}^T\boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b} \\ & \boldsymbol{x} \geq \mathbf{0} \end{aligned}$$	$$\begin{aligned} \max_{\boldsymbol{y}} \quad & \boldsymbol{b}^T\boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^T\boldsymbol{y} \leq \boldsymbol{c} \\ & \boldsymbol{y} \geq \mathbf{0} \end{aligned}$$

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Manufacturing problem

Suppose company A can manufacture n products from m materials.

- Manufacturing one unit of product i needs a_{ij} units of material j
- The unit price of product i is b_i
- The inventory of material j is c_j

where $a_{ij} > 0$, $b_i > 0$, $c_j > 0$.

Let y_i denote the amount of product i manufactured. To maximize its revenue, the company solves the following LP,

$$\begin{aligned} \max_{\boldsymbol{y}} \quad & \sum_{i=1}^{n} b_i y_i \\ \text{s.t.} \quad & \sum_{i=1}^{n} y_i a_{ij} \leq c_j, \forall j = 1, 2, \ldots, m \\ & y_i \geq 0, \quad i = 1, 2, \ldots, n \end{aligned}$$

Manufacturing problem (cont'd)

Now suppose company B offers to buy the raw materials at the price of x_j per unit of material j.

The equivalent offer for one unit of product i is $\sum_{j=1}^{m} a_{ij}x_j$. Company A will accept the offer only if

$$\sum_{j=1}^{m} a_{ij}x_j \geq b_i, \quad i = 1, 2, \ldots, n$$

To minimizes its cost, company B solves the dual LP,

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \sum_{j=1}^{m} c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{m} a_{ij}x_j \geq b_i, \forall i = 1, 2, \ldots, n \\ & x_j \geq 0, \quad j = 1, 2, \ldots, m \end{aligned}$$

Optimal transport problem

Recall the optimal transport problem in §1,

$$
\begin{array}{lll}
\min\limits_{(x_{ij})} & \displaystyle\sum_{i=1}^{n}\sum_{j=1}^{m} c_{ij}x_{ij} & \\
\text{s. t.} & \displaystyle\sum_{i=1}^{n} x_{ij} = b_j & \text{for} \quad j = 1, 2, \ldots, m \\
& \displaystyle\sum_{j=1}^{m} x_{ij} \le a_i & \text{for} \quad i = 1, 2, \ldots, n \\
& x_{ij} \ge 0 & \text{for} \quad i = 1, 2, \ldots, n;\ j = 1, 2, \ldots, m
\end{array}
$$

- x_{ij}: quantity shipped from warehouse i to customer j
- c_{ij}: unit shipping cost from warehouse i to customer j
- a_i: inventory at warehouse i
- b_j: demand at customer j

Optimal transport problem (cont'd)

Now instead of actually shipping the products, you decide to fulfill the orders by trading with another seller of the same product, who

- buys your stock at warehouse i at the unit price μ_i
- delivers the order of customer j at the unit price λ_j

The cost of "sending" one unit from i to j is $\lambda_j - \mu_i$. The deal will be competitive if

$$\lambda_j - \mu_i \leq c_{ij}$$

To maximize his profit, the other seller solves the dual LP,

$$\begin{array}{lll} \max\limits_{(\lambda_j),(\mu_i)} & \displaystyle\sum_{j=1}^{m} b_j \lambda_j - \sum_{i=1}^{n} a_i \mu_i & \\ \text{s.t.} & \lambda_j - \mu_i \leq c_{ij} & \text{for} \quad i = 1, 2, \ldots, n;\ j = 1, 2, \ldots, m \\ & \mu_i \geq 0 & \text{for} \quad i = 1, 2, \ldots, n \end{array}$$

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Weak and strong duality

Consider an LP and its dual. WLOG, consider the inequality form,

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \boldsymbol{c}^T\boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b} \end{aligned} \qquad \text{(P)}$$

$$\begin{aligned} \max_{\boldsymbol{y}} \quad & \boldsymbol{b}^T\boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^T\boldsymbol{y} = \boldsymbol{c} \\ & \boldsymbol{y} \geq \boldsymbol{0} \end{aligned} \qquad \text{(D)}$$

Weak duality. If $\boldsymbol{x}$ is primal feasible and $\boldsymbol{y}$ is dual feasible, then

$$\boldsymbol{c}^T\boldsymbol{x} \geq \boldsymbol{b}^T\boldsymbol{y}$$

Proof.

$$\boldsymbol{c}^T\boldsymbol{x} = (\boldsymbol{A}^T\boldsymbol{y})^T\boldsymbol{x} = \boldsymbol{y}^T\boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{y}^T\boldsymbol{b}$$

Strong duality. If either (P) or (D) has a finite optimal value, so does the other, the optimal solutions $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$ exist, and $\boldsymbol{c}^T\boldsymbol{x}^* = \boldsymbol{b}^T\boldsymbol{y}^*$.

Proof. Will follow from Slater's Theorem. Show weaker form next.

Strong duality

Lemma. If all (equality and inequality) constraint functions are affine, then the KKT conditions hold at a local minimum.

Note. We do not assume the local minimum is a regular point.

Theorem. If (P) has an optimal solution, then (D) also has an optimal solution and strong duality holds.

Proof. Let $\boldsymbol{x}^*$ be the optimal solution of (P). By the KKT conditions, there exist Lagrange multipliers $\boldsymbol{\mu}^*$ s.t.

1. $\boldsymbol{c} - \boldsymbol{A}^T \boldsymbol{\mu}^* = \boldsymbol{0}$ (stationarity)
2. $\boldsymbol{\mu}^* \geq \boldsymbol{0}$ (nonnegativity)
3. $(\boldsymbol{\mu}^*)^T (\boldsymbol{A}\boldsymbol{x}^* - \boldsymbol{b}) = 0$ (complementary slackness)

By 1 and 2, $\boldsymbol{\mu}^*$ is feasible for (D). By 1 and 3,

$$\boldsymbol{c}^T \boldsymbol{x}^* = (\boldsymbol{A}\boldsymbol{\mu}^*)^T \boldsymbol{x}^* = \boldsymbol{b}^T \boldsymbol{\mu}^*$$

By weak duality, $\boldsymbol{\mu}^*$ is optimal for (D) and hence strong duality holds.

Example

Recall the following pair of primal and dual LP,

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f(\boldsymbol{x}) = x_1 + 2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 2 \\ & \boldsymbol{x} \geq \mathbf{0} \end{aligned}$$

$$\begin{aligned} \max_{\boldsymbol{\mu}\in\mathbb{R}^3} \quad & \psi(\boldsymbol{\mu}) = 2\mu_1 \\ \text{s.t.} \quad & 2\mu_1 + \mu_2 = 1 \\ & \mu_1 + \mu_3 = 2 \\ & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

- The primal LP can be solved graphically, $\boldsymbol{x}^* = (1, 0)$, $f^* = 1$.
- The dual is equivalent to

$$\begin{aligned} \max_{\mu_1} \quad & 2\mu_1 \\ \text{s.t.} \quad & 2\mu_1 \leq 1 \\ & \mu_1 \leq 2 \\ & \mu_1 \geq 0 \end{aligned}$$

So $\boldsymbol{\mu}^* = (\frac{1}{2}, 0, \frac{3}{2})$, $\psi^* = 1 = f^*$.

Wasserstein distance

In the optimal transport problem, let $\boldsymbol{x}_i$ be the location of warehouse i, and $\boldsymbol{y}_j$ be the location of customer j. Suppose

$$\sum_{i=1}^{n} a_i = 1 = \sum_{j=1}^{m} b_j$$

Note a, b can be interpreted as two discrete probability distributions,

$$\Pr[\boldsymbol{X} = \boldsymbol{x}_i] = a_i, \quad \Pr[\boldsymbol{Y} = \boldsymbol{y}_j] = b_j$$

Now suppose the shipping cost c_{ij} is determined by the distance between $\boldsymbol{x}_i$ and $\boldsymbol{y}_j$,

$$c_{ij} = d(\boldsymbol{x}_i, \boldsymbol{y}_j)$$

e.g. $d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_2$.

Wasserstein distance (cont'd)

The (first) Wasserstein distance $W_1(a,b)$ between the two distributions a and b is the optimal value of the optimal transport problem.

$$\begin{aligned}
\min_{(x_{ij})} \quad & \sum_{i=1}^{n}\sum_{j=1}^{m} c_{ij}x_{ij} \\
\text{s. t.} \quad & \sum_{i=1}^{n} x_{ij} = b_j \qquad \text{for} \quad j = 1,2,\ldots,m \\
& \sum_{j=1}^{m} x_{ij} = a_i \qquad \text{for} \quad i = 1,2,\ldots,n \\
& x_{ij} \geq 0 \qquad \text{for} \quad i = 1,2,\ldots,n;\ j = 1,2,\ldots,m
\end{aligned}$$

Note the feasible set is $\Pi(a,b)$, the set of all joint distributions with marginals a and b,

$$W_1(a,b) = \inf_{x\in\Pi(a,b)} \sum_{i,j} d(\boldsymbol{x}_i,\boldsymbol{y}_j)x_{ij} = \inf_{x\in\Pi(a,b)} \mathbb{E}_{(\boldsymbol{X},\boldsymbol{Y})\sim x}[d(\boldsymbol{X},\boldsymbol{Y})]$$

Wasserstein distance (cont'd)

The dual problem is

$$\begin{aligned} \max_{(\lambda_j),(\mu_i)} \quad & \sum_{j=1}^{m} b_j \lambda_j - \sum_{i=1}^{n} a_i \mu_i \\ \text{s. t.} \quad & \lambda_j - \mu_i \le c_{ij} \qquad \text{for} \quad i = 1, 2, \ldots, n;\ j = 1, 2, \ldots, m \end{aligned}$$

Note. We do not have the constraints $\mu_i \ge 0$ as on slide 12 due to the modification in the primal, "$\le a_i$" $\to$ "$= a_i$".

By strong duality, $W_1(a, b)$ is also equal to the dual optimal value.

- $x_{ij} = a_i b_j$ is primal feasible (what's the probabilistic interpretation?)
- the optimal cost is lower bounded by $\min_{ij} c_{ij}$ and hence finite.

Rewriting $\lambda_j = \lambda(\boldsymbol{y}_j)$ and $\mu_i = \mu(\boldsymbol{x}_i)$,

$$W_1(a, b) = \sup_{\lambda(\boldsymbol{y}_j) - \mu(\boldsymbol{x}_i) \le d(\boldsymbol{x}_i, \boldsymbol{y}_j),\ \forall i,j} \left[\sum_{j=1}^{m} b_j \lambda(\boldsymbol{y}_j) - \sum_{i=1}^{n} a_i \mu(\boldsymbol{x}_i) \right]$$

Wasserstein distance (cont'd)

We can actually restrict λ and μ to be the same function. Then the constraints become

$$h(\boldsymbol{y}_j) - h(\boldsymbol{x}_i) \leq d(\boldsymbol{x}_i, \boldsymbol{y}_j), \quad \forall i, j$$

which can be further strengthened to 1-Lipschitz continuity of h, i.e.

$$h(\boldsymbol{z}) - h(\boldsymbol{z}') \leq d(\boldsymbol{z}, \boldsymbol{z}'), \quad \forall \boldsymbol{z}, \boldsymbol{z}'$$

Theorem (Kantorovich-Rubinstein duality).

$$W_1(a, b) = \sup_{h \text{ is } 1\text{-Lipschitz}} \left[\sum_{j=1}^{m} b_j h(\boldsymbol{y}_j) - \sum_{i=1}^{n} a_i h(\boldsymbol{x}_i) \right]$$

In the general case,

$$\begin{aligned} W_1(a, b) &= \inf_{x \in \Pi(a,b)} \mathbb{E}_{(\boldsymbol{X},\boldsymbol{Y}) \sim x}[d(\boldsymbol{X}, \boldsymbol{Y})] \\ &= \sup_{h \text{ is } 1\text{-Lipschitz}} \mathbb{E}_{\boldsymbol{Y} \sim b}[h(\boldsymbol{Y})] - \mathbb{E}_{\boldsymbol{X} \sim a}[h(\boldsymbol{X})] \end{aligned}$$

Appendix: Proof of discrete case

Given μ, define

$$h(\boldsymbol{z}) = \min_k [\mu(\boldsymbol{x}_k) + d(\boldsymbol{x}_k, \boldsymbol{z})]$$

Then h is 1-Lipschitz. Indeed, by the triangle inequality for d,

$$\begin{aligned} h(\boldsymbol{z}) &= \min_k [\mu(\boldsymbol{x}_k) + d(\boldsymbol{x}_k, \boldsymbol{z})] \\ &\leq \min_k \{\mu(\boldsymbol{x}_k) + d(\boldsymbol{x}_k, \boldsymbol{z}') + d(\boldsymbol{z}, \boldsymbol{z}')\} = h(\boldsymbol{z}') + d(\boldsymbol{z}, \boldsymbol{z}') \end{aligned}$$

If λ and μ satisfy the dual constraints, then replacing both λ and μ by h yields a solution that is at least as good as the old one,

- feasibility follows from 1-Lipschitz continuity of h,

$$h(\boldsymbol{y}_j) - h(\boldsymbol{x}_i) \leq d(\boldsymbol{x}_i, \boldsymbol{y}_j), \quad \forall i, j$$

- improvement follows from the dual constraints and the definition of h,

$$\lambda(\boldsymbol{y}_j) \leq \min_k [\mu(\boldsymbol{x}_k) + d(\boldsymbol{x}_k, \boldsymbol{y}_j)] = h(\boldsymbol{y}_j), \quad \forall j$$

$$\mu(\boldsymbol{x}_i) = \mu(\boldsymbol{x}_i) + d(\boldsymbol{x}_i, \boldsymbol{x}_i) \geq \min_k [\mu(\boldsymbol{x}_k) + d(\boldsymbol{x}_k, \boldsymbol{x}_i)] = h(\boldsymbol{x}_i), \quad \forall i$$

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Dual LP via Lagrangian

The Lagrangian for the general LP on slide 6 is

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{c}^T \boldsymbol{x} - \boldsymbol{\lambda}^T (\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) - \boldsymbol{\mu}^T (\boldsymbol{G}\boldsymbol{x} - \boldsymbol{h})$$

If $\boldsymbol{\mu} \geq \boldsymbol{0}$ and $\boldsymbol{x} \in X$, i.e. $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and $\boldsymbol{G}\boldsymbol{x} \geq \boldsymbol{h}$, then

$$f(\boldsymbol{x}) = \boldsymbol{c}^T \boldsymbol{x} \geq \boldsymbol{c}^T \boldsymbol{x} - \underbrace{\boldsymbol{\lambda}^T (\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b})}_{=0} - \underbrace{\boldsymbol{\mu}^T (\boldsymbol{G}\boldsymbol{x} - \boldsymbol{h})}_{\geq 0} = \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$$

Taking the infimum over $\boldsymbol{x} \in X$ first and then relaxing the constraint,

$$f^* = \inf_{x \in X} f(\boldsymbol{x}) \geq \inf_{\boldsymbol{x} \in X} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \geq \inf_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) =: \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

To maximize the lower bound, solve the following dual problem

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \boldsymbol{0} \end{aligned}$$

Dual LP via Lagrangian (cont'd)

Note

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = (\boldsymbol{c} - \boldsymbol{A}^T\boldsymbol{\lambda} - \boldsymbol{G}^T\boldsymbol{\mu})^T\boldsymbol{x} + \boldsymbol{b}^T\boldsymbol{\lambda} + \boldsymbol{h}^T\boldsymbol{\mu}.$$

An affine function is bounded below iff the coefficient for $\boldsymbol{x}$ is zero[1].

The dual problem

$$\begin{aligned} \max_{\boldsymbol{\lambda},\boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{cases} \boldsymbol{b}^T\boldsymbol{\lambda} + \boldsymbol{h}^T\boldsymbol{\mu}, & \text{if } \boldsymbol{c} - \boldsymbol{A}^T\boldsymbol{\lambda} - \boldsymbol{G}^T\boldsymbol{\mu} = \boldsymbol{0} \\ -\infty & \text{otherwise} \end{cases} \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \boldsymbol{0} \end{aligned}$$

which is equivalent to the dual LP

$$\begin{aligned} \max_{\boldsymbol{\lambda},\boldsymbol{\mu}} \quad & \psi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{b}^T\boldsymbol{\lambda} + \boldsymbol{h}^T\boldsymbol{\mu} \\ \text{s.t.} \quad & \boldsymbol{A}^T\boldsymbol{\lambda} + \boldsymbol{G}^T\boldsymbol{\mu} = \boldsymbol{c} \\ & \boldsymbol{\mu} \geq \boldsymbol{0} \end{aligned}$$

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[1] Consider $f(\boldsymbol{x}) = \boldsymbol{a}^T\boldsymbol{x} + c$. If $\boldsymbol{a} = \boldsymbol{0}$, then $\inf_{\boldsymbol{x}} f(\boldsymbol{x}) = c$. If $\boldsymbol{a} \neq \boldsymbol{0}$, letting $\boldsymbol{x} = -t\boldsymbol{a}$ and $t \to +\infty$ yields $\inf_{\boldsymbol{x}} f(\boldsymbol{x}) \leq -t\|\boldsymbol{a}\|^2 + c \to -\infty$.